PROJECT 07: Retail Billing with Multiple Discount Rules

1. Problem Statement

Generate an itemized invoice for multiple products, quantities, category-specific taxes and threshold discounts, subject to a maximum discount cap.

2. Objectives

• Calculate line totals and subtotal.
• Apply a threshold discount and cap it at a maximum value.
• Calculate category-specific tax.
• Print a clear invoice and grand total.

3. Assumptions

• A 10% basket discount applies when subtotal is at least Rs.1,000.
• Total discount cannot exceed Rs.500.
• Each product has a supplied tax rate; tax is calculated on line value before discount for this sample policy.
• Product prices and quantities must be non-negative; quantities are whole numbers.

4. Algorithm

1. Read product name, price, quantity and tax rate.
2. Calculate each line amount and subtotal.
3. Apply 10% discount if subtotal reaches Rs.1,000, capped at Rs.500.
4. Calculate tax using each product's tax rate.
5. Add taxes and subtract discount to get grand total.
6. Print the itemized invoice.

5. Flowchart

START
↓
Input data and validate
↓
Process rules and calculations
↓
Check conditions / classify results
↓
Display result and END

6. Pseudocode

```
START
  subtotal = 0
  tax_total = 0
  FOR each item
    line = price * quantity
    subtotal = subtotal + line
    tax_total = tax_total + line * tax_rate
    DISPLAY line
  IF subtotal >= 1000: discount = min(500, subtotal * 10%)
  ELSE: discount = 0
  grand_total = subtotal - discount + tax_total
  DISPLAY subtotal, discount, tax_total, grand_total
END
```

7. Python Code

```
items = [
    {"name": "Notebook", "category": "stationery", "price": 80, "qty": 5, "tax": 0.05},
    {"name": "Headphones", "category": "electronics", "price": 1200, "qty": 1, "tax": 0.18},
    {"name": "Pen", "category": "stationery", "price": 20, "qty": 10, "tax": 0.05},
]
subtotal = sum(i["price"] * i["qty"] for i in items)
```

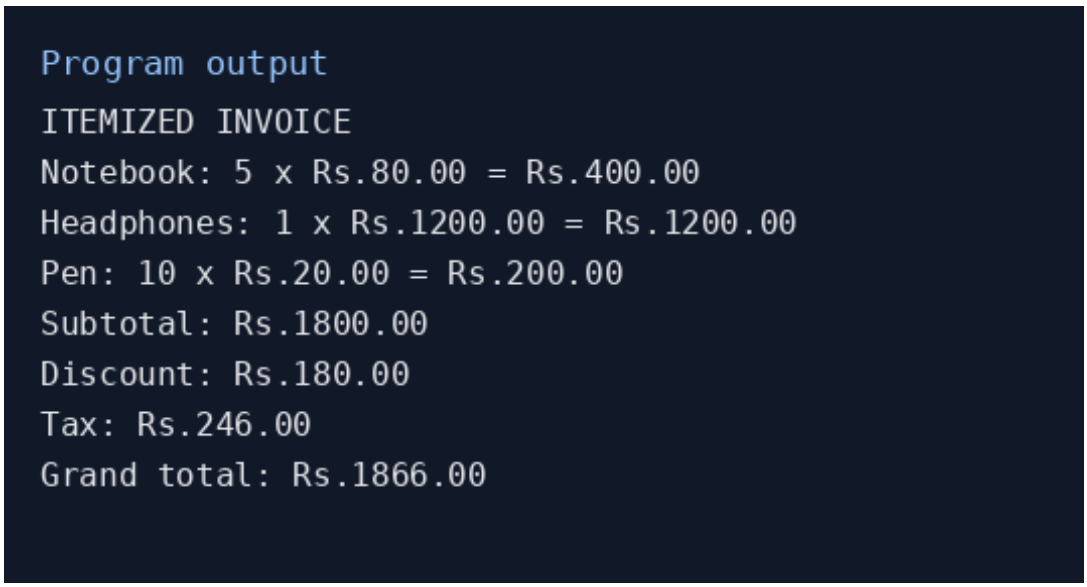
```
discount = min(500, subtotal * 0.10) if subtotal >= 1000 else 0
tax_total = sum(i["price"] * i["qty"] * i["tax"] for i in items)
print("ITEMIZED INVOICE")
for i in items:
    amount = i["price"] * i["qty"]
    print(f"{i['name']}: {i['qty']} x Rs.{i['price']:.2f} = Rs.{amount:.2f}")
grand_total = subtotal - discount + tax_total
print(f"Subtotal: Rs.{subtotal:.2f}")
print(f"Discount: Rs.{discount:.2f}")
print(f"Tax: Rs.{tax_total:.2f}")
print(f"Grand total: Rs.{grand_total:.2f}")
```

8. Test Cases

Test	Input / condition	Expected result
Below threshold	Subtotal Rs.500	No threshold discount
At threshold	Subtotal Rs.1,000	10% discount = Rs.100
Discount cap	Subtotal Rs.8,000	Discount capped at Rs.500
Mixed taxes	Stationery 5%, electronics 18%	Tax calculated per item category

9. Output Screenshot

The following screenshot-style image was generated from the captured output of the program execution.



Captured output

```
ITEMIZED INVOICE
Notebook: 5 x Rs.80.00 = Rs.400.00
Headphones: 1 x Rs.1200.00 = Rs.1200.00
Pen: 10 x Rs.20.00 = Rs.200.00
Subtotal: Rs.1800.00
Discount: Rs.180.00
Tax: Rs.246.00
Grand total: Rs.1866.00
```

10. Conclusion

The program creates an auditable itemized invoice and combines threshold discount logic with item-level tax rates.

Academic note: The program was executed with the sample data shown in the source code. Policies, rates, thresholds and exchange rates are illustrative and should be adjusted to the assignment's official requirements.